Module 08: Planning in Robotics

Learning Objectives

1. Define **Planning** within the context of Robotics.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Robotics, we see Planning manifest in: * **Specific Example 1**: [Add domain-specific example here] * **Specific Example 2**: [Add domain-specific example here]

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.